

B5 tier assignment to PINNED-CLOSURE: the beyond-layer bound closed within the amended-class second-cumulant scope

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

Operator verdict #14 set B5 = T5-CANDIDATE; this note records the operator-authorized assignment to T5 PINNED-CLOSURE. The mandatory TSv2 scope string is

CLOSED@H-ADM-COH-AMENDED-CLASS (second-cumulant order).

The closure: within the H-ADM-COH amended competitor class, at matched second-cumulant order, the beyond-layer off-diagonal correction is bounded below the layer margin for every admissible pattern (STEP-5B gate, CLOSED-CONDITIONAL, margins $\times 59.4 / \times 8.8 / \times 2.6$).

2. Why T5 (PINNED-CLOSURE), not T4 and not T6

Above T4 (STRONG-EVIDENCE). T4 is “numerics + analytic bounds, not a theorem.” B5 is past that: the gate STEP-5B was flipped to CLOSED-CONDITIONAL by operator verdict #14; the closure rests on theorem-grade pillars — the P^2 -representation floor (R-001, operator monotonicity), the operator-derived $c_R = 4\sqrt{14}$, the universal single-circle theorem (R-002, $K = 14$ sharp), the antipodal-carrier partition (R-003) — assembled into a closure, not a collection of evidence.

Below T6 (CONDITIONAL-THEOREM). T6 is a clean theorem under named hypotheses. B5 is not yet that: the assembled chain contains strong-evidence-grade links (the coherence indistinguishability lemma AddC is a T4 analytic-skeleton-plus-machine result; the cross-reading constant was machine-bounded then analytically pinned in AddE); the endpoint margin is thin ($\times 2.6$); the scope is second-cumulant with the third-cumulant endpoint marginal (SC-SCOPE). A closure resting on a pinned operational scope with mixed-grade internal links is exactly PINNED-CLOSURE.

Hence T5, scope pinned to the amended class and second-cumulant order, with the reproduction package below.

3. Reproduction package (TSv2 T5 requirement)

Artefact	Status
codes/vacuum/beyond_layer_gershgorin_bound.py	192/192 self-test asserts PASS
v1.14.0	
run artefact	present
claims/B5-BEYOND-LAYER-BOUND/runs/260605-gershgorin-reduction/result.json	
note chain (main v2.0; AddA v1.3; AddB; AddC; AddD; AddE)	FORM-CHECK PASS, PDFs beside source
development lineage	generated, ordered
claims/B5-BEYOND-LAYER-BOUND/LINEAGE.md	

Reproduction command: `python codes/vacuum/beyond_layer_gershgorin_bound.py` $\Rightarrow$ claims 192/192 PASS; the closure margins $\times 59.4 / \times 8.8 / \times 2.6$ are emitted across the anchor band.

4. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3.5; binding gate, no external review this turn)

- (α) “The closure rests on *H-ADM-COH*, an operator- *ADOPTED* class amendment, not a proven fact — is a closure that depends on a class choice really *T5*?” ADDRESSED by the scope string: *T5 PINNED-CLOSURE* explicitly admits an operational scope pin (*CLOSED@...*), and *H-ADM-COH* is that pin, named in the scope and registered in *GATES.md*. The closure is honest within the named class; the unrestricted class is the *DR-2* route (research-grade), not claimed here.
- (β) “The endpoint margin is $\times 2.6$ and the third-cumulant endpoint is marginal — is the closure solid?” VALID-with-mitigation: the $\times 2.6$ floor is theorem-grade within second-cumulant (AddE); the third-cumulant marginality is the *SC-SCOPE* pin (second-cumulant order is in the scope string). The all-orders endpoint is an open lift, not part of the *T5* claim.
- (γ) “Does *B5* depend on the provisional 20/9 incidence route?” DISMISSED: the closure path uses the operator- derived $c_R = 4\sqrt{14}$ (theorem-grade) and the admissibility route; the 20/9 route is a supporting candidate (constant unpinned), explicitly EXCLUDED from the closure path (AddA v1.3). No *T5* figure depends on it.
- (δ) “Reproduction package?” SATISFIED: script 192/192 + run artefact + the note chain (section 3).
- (ϵ) “Tier monotonicity: *B1* (*T6*) uses *B5* — must *B5* be $\geq T6$?” DISMISSED: *B1* does not hard-depend on *B5*; *B1* carries *B5*’s closure as the NAMED hypothesis *H-ADM-COH* (a *T6* claim may rest on a sub-*T6* input promoted to a named hypothesis, TSv2 §4.4). The DAG is acyclic. *B5* at *T5* is consistent with *B1* at *T6*.

No objection is UPHOLD as a blocker; one DISMISSED-via-scope, one VALID-with-mitigation (the second-cumulant/endpoint pin, in the scope string), three DISMISSED.

5. Result footer

Result ID: B5-BEYOND-LAYER-BOUND (beyond-layer bound)

Precise statement:	Within the H-ADM-COH amended competitor class at second-cumulant order, the beyond-layer off-diagonal correction is bounded below the layer margin for every admissible pattern (STEP-5B CLOSED-CONDITIONAL; floors $\times 59.4/\times 8.8/\times 2.6$). Closure pillars: P^2 floor (R-001), $c_R = 4 \sqrt{14}$, single-circle $K=14$ (R-002), carrier partition (R-003), H-ADM-COH admissibility, indistinguishability lemma.
Scope (mandatory T5):	CLOSED@H-ADM-COH-AMENDED-CLASS, second-cumulant order. NOT unrestricted (DR-2 research-grade); NOT all-orders (SC-SCOPE).
Dependencies:	A1-KERNEL-CONV (convention); B2-PROPA-HLAYER (layer margin). Soft: B1 consumes this via H-ADM-COH.
Evidence grade:	ANALYTIC (closure pillars) + EXECUTED (192/192)
Reproduction command:	python codes/vacuum/beyond_layer_gershgorin_bound.py
Expected output:	claims 192/192 PASS; margins $\times 59.4/\times 8.8/\times 2.6$.
Falsification gate:	An admissible amended-class pattern with beyond-layer correction exceeding the margin; a section breaking the P^2 floor; loss of the H-ADM-COH class definition.
Tier before / after:	T4 (T5-CANDIDATE) / T5 PINNED-CLOSURE (operator-authorized "proceed with item 2", 2026-06-06).
No-overclaim statement:	T5 is closure WITHIN the pinned amended-class second-cumulant scope, resting on a chain with some T4-grade internal lemmas (hence not T6) and not unconditional (hence not T7). The unrestricted-class result is DR-2 (research-grade); the all-orders lift is open (SC-SCOPE).
Next required action:	Promotion to T6 would require discharging the T4-grade lemmas to theorem grade and a clean conditional-theorem statement; the unrestricted T7 path is DR-2. Standing decision H-A0 -> H-ANCHOR remains.